MILITARY SPECIFICATION

VALVE CAP, VALVE CORE, VALVE SPUD, AND VALVE, PNEUMATIC TIRE: REPAIR AND REPLACEMENT, FOR GROUND VEHICLES

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope. This specification covers valve caps, valve cores, valve spuds, and repair and replacement valve assemblies for tubeless tires or inner tubes used on military ground vehicles. Valve assemblies include valve caps, valve cores, and valve spuds (when applicable, see 6.1).

1.2 Classification. Pneumatic-tire valves, spuds, caps, and cores shall be of the following types and classes as specified (see 6.2).

Type I—Valve assemblies, tube.

Class 1 —Tube valve, replacement, air use only, rubber-covered stem and base, standard bore, straight.

Class 2 —Tube valve, replacement, air use only, rubber-covered base, bare metal stem with body threads, standard bore, straight.

Class 3 —Tube valve, replacement, air use only, rubber-covered base, bare metal stem with body threads, standard bore, bent.

Class 4 —Tube valve replacement, air use only, rubber-covered base, bare metal stem without body threads standard bore, straight.

Class 5 —Tube valve, replacement, air use only, rubber-covered base, bare metal stem without body threads, standard bore, 40° bend.

Class 6 —Tube valve, replacement, air use only, rubber-covered base, bare metal stem without body threads, standard bore, 90° bend.

Class 7 —Tube valve, replacement, air-water use, rubber-covered stem and base, standard bore, straight.

Class 8 —Tube valve, replacement, air-water use, rubber-covered base, bare metal stem, standard bore, straight.

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Class 9 —Tube valve, repair, screw-on, standard bore, straight.
Class 10 —Tube valve, repair, screw-on, standard bore, bent.
Class 11 —Tube valve, repair, screw-on, large bore, straight.
Class 12 —Tube valve, repair, screw-on, large bore, bent.

Type II—Valve assemblies, tubeless.
Class 1 —Tubeless valve, replacement, snap-in, standard bore.
Class 2 —Tubeless valve, replacement, clamp-in, truck, for drop-center rims and side-of-well, valve-hole location, standard bore, straight.
Class 3 —Tubeless valve, replacement, clamp-in, truck, for sealed rims and center-of-base, valve-hole location, standard bore, bent.
Class 4 —Tubeless valve, replacement, clamp-in, off-the-road, large bore, air-water use, with removable core housing, straight.
Class 5 —Tubeless valve, replacement, clamp-in, off-the-road, large bore, air-water use, with removable core housing, bent.
Class 6 —Tubeless valve, replacement, clamp-in, off-the-road, standard bore, air use only, straight.
Class 7 —Tubeless valve, replacement, clamp-in, off-the-road, standard bore, air use only, bent.

Type III—Valve spuds.
Class 1 —Spud, valve, tubed tire, repair, large bore, straight.
Class 2 —Spud, valve, tubeless tire, repair, large bore, straight.

Type IV—Valve caps.
Class 1 —Valve cap, standard bore.
Class 2 —Valve cap, large bore.

Type V—Valve cores.
Class 1 —Valve core, standard bore.
Class 2 —Valve core, large bore.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ–B–626 —Brass, Leaded and Nonleaded; Rods, Shapes, Forgings, and Flat Products With Finished Edge (Bars, Flat Wire and Strips).
QQ–W–390 —Wire: Nickel-Chromium-Iron Alloy.
ZZ–I–550 —Inner Tube, Pneumatic Tire (Automobiles, Trucks, Motorcycles, and Other Ground Vehicles).
PPP–B–585 —Boxes; Wood, Wirebound.
PPP–B–591 —Boxes; Fiberboard, Wood-Cleated.
PPP–B–601 —Boxes, Wood, Cleated-Plywood.

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PPP–B–621 —Boxes, Wood, Nailed and Lock Corner.
PPP–B–636 —Boxes, Fiber.
PPP–C–843 —Cushioning Material, Cellulosic.
PPP–P–291 —Paperboard, Wrapping, Cushioning.

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JAN–P–100 —Packaging and Packing for Overseas Shipment General Specification.
MIL–P–116 —Preservation, Method of.
MIL–B–138 —Boxes, Wood, Fiberboard-Lined, for Overseas Shipment. (For Weights of Contents Not Exceeding 500 Pounds).
MIL–R–6855 —Rubber, Synthetic, Sheet, Molded and Extruded, for Aircraft Application.

STANDARDS

FEDERAL

FEDERAL TEST METHOD STD. NO. 601—Rubber: Sampling and Testing.

MILITARY

MIL–STD–105 —Sampling Procedures and Tables for Inspection by Attributes.
MIL–STD–129 —Marking for Shipment and Storage.
MIL–STD–130 —Identification Marking of U. S. Military Property.
MIL–STD–417 —Rubber Compositions, Vulcanized, General Purpose, Solid (Symbols and Tests).

For applicable MS standards covered by this specification, see supplement 1.

2.2 **Other publications.** The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN TRUCKING ASSOCIATIONS INC.
National Motor Freight Classification
(Application for copies should be addressed to the American Trucking Associations Inc., 1424 Sixteenth Street, N.W., Washington 6, D.C.)

CONSOLIDATED CLASSIFICATION COMMITTEE
Consolidated Freight Classification
(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

THE TIRE AND RIM ASSOCIATION, INC.
Year Book
(Application for copies should be addressed to The Tire and Rim Association, Inc., 2001 First National Tower, Akron 8, Ohio.)

3. REQUIREMENTS

3.1 **Preproduction sample.** The valves, spuds, cores, and caps furnished under this specification shall be products which have been represented by preproduction samples that have been inspected and have passed the preproduction testing specified herein (see 4.2 and 6.2).

3.2 **Materials.**

3.2.1 *Valve stems.* Valve stems shall be of brass as specified in 3.4.10.

3.2.2 *Rubber covers and bases.* Rubber covers and bases of tire valves for application to inner tubes conforming the Specification ZZ–I–550 shall be made from IIR (butyl) or natural rubber conforming to compound R 615 SB of Standard MIL–STD–417 (Ord). Rubber covers for snap-in valves of tubeless tires shall be made from SBR or natural rubber conforming to compound R 715 SB of Standard MIL–STD–417 (Ord), unless otherwise specified (see 6.2).

3.2.2.1 *Base preparation.* Tire valves with rubber bases shall have a layer of uncured gum stock cemented to the bases to permit their airtight application to the buffed and cleaned valve-hole locations on the inner tubes when cured with commercially-available field vulcanizers at approximately 300° F, the curing time for valve replacements on

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passanger-car tubes being 12 minutes and on truck and large-vehicle tubes 15 to 18 minutes.

3.2.3 *Valve caps.* Valve cap shells and swivel seat plates shall be of brass conforming to Specification QQ–B–626, composition 22 half hard, or other suitable metal. Valve cap seats shall be of rubber conforming to Standard MIL–STD–417, Grade SC 615 or other material meeting the performance requirements of this specification. Plastic shall not be considered acceptable (see 3.3.3).

3.2.4 *Valve cores.* Valve core barrels, bridges, stems, and gasket cups shall be: brass conforming to Specification QQ–B–626, composition 22 half hard; seal and cup gasket of synthetic rubber conforming to specification MIL–R–6855, class II, grade 60 or 80; spring of stainless steel conforming to Specification QQ–W–390, class 2; or equivalent materials. Material characteristics shall not change sufficiently to affect valve conformance to 3.4.4 when given the heat-leakage test as specified in 4.5.1 or 4.5.2 (see 3.3.4).

3.2.5 *Metal valve bases and spuds.* Metal valve bases and spuds shall be of brass conforming to Specification QQ–B–626, composition 22 half hard, or equivalent material.

3.3 Valve design and construction.

3.3.1 *Assembly.* Each valve (see MS51357 through MS51373) shall be furnished as a complete assembly, including a metal valve stem, rubber covered if specified (see 6.1), a metal valve cap (see 3.2.3), a valve core (see 3.2.4) and, when required, a valve spud (see MS51374). Type I, classes 11 and 12 valve assemblies shall be as shown in the year book of the Tire and Rim Association Inc.

3.3.2 *Parts interchangeability.* Component parts of valve assemblies used for in-the-field replacement shall be interchangeable with the original parts without modification.

3.3.3 *Valve-cap construction.* Valve cap design and construction shall conform to Military Standards MS51375 and MS51376.

3.3.4 *Valve core construction.* Valve cores shall be designed and constructed in conformance to Military Standards MS51377 and MS MS51378.

3.3.5 *Valve spud construction.* Valve spuds shall be designed and constructed in conformance to Military Standard MS51374.

3.4 Physical properties.

3.4.1 *Pull valves.* Valves with rubber covers or bases shall be pull-tested as specified in 4.5.3 to determine conformance of their pull values to those in table I.

TABLE I. *Minimum pull values.*

TR valve No.	Minimum pull value in pounds	Adapter ring diameter in inches
13	100	1¼
13CW	100	1¼
15	100	1¼
15CW	100	1¼
25	100	1¼
75A	300	1¼
77A	300	1¼
78A	300	1¼
87	100	1¼
88	100	1¼
135	100	1¼
150	100	1¼
150CW	100	1¼
175A	300	1¼
177A	300	1¼
179A	300	1¼
218	100	1¼
220	100	1¼

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3.4.2 *Durometer hardness (shore A).* The durometer hardness of the valve-rubber compound (see 4.2.1) shall be within the limits of 55 to 65 inclusive for natural and butyl synthetic rubber and within the limits of 65 to 75 inclusive for SBR when tested as specified in 4.5.4.

3.4.3 *Valve-stem base closure.* When tested as specified in 4.5.5, visual examination shall reveal sufficient spinning or closing of the inner wall of the metal groove, without wall cracking, to effect a mechanical lock of the rubber in the groove of type I, classes 1 through 6 valves.

3.4.4 *Valve core leakage.* Valve stems, with cores tightened to a torque of 2 pound-inches, shall show no leakage when tested as specified in 4.5.6. An air bubble which forms on the mouth of the valve and breaks loose within a two-minute period shall constitute leakage except in the case of a valve being tested for leakage at low temperature (see 4.5.1.2.2 and 4.5.2.2), an air bubble forming and breaking loose within a one-minute period shall constitute leakage.

3.4.5 *Temperature - range performance (type I, all classes, and type II, classes 2 through 7).*

3.4.5.1 *Leakage at high temperature.* Valves shall conform to the leakage requirements of 3.4.4 when tested as specified in 4.5.1.2.1 in a water bath of 200° F. after storage for 24 hours at 212° F.

3.4.5.2 *Leakage at low temperature.* Valves shall conform to the leakage requirements of 3.4.4 when tested as specified in 4.5.1.2.2 while in an alcohol bath at minus 65° F.

3.4.6 *Temperature performance (type II, class 1).*

3.4.6.1 *Leakage at high temperature.* Valves shall conform to the leakage requirements of 3.4.4 when tested as specified in 4.5.2.1 at 212° F.

3.4.6.2 *Leakage at low temperature.* Valves shall conform to the leakage requirements of 3.4.4 when tested as specified in 4.5.2.2 at minus 65° F.

3.4.7 *Valve cap leakage.* Valve caps shall seal on valve stems and show no visible air leakage when tested as specified in 4.5.9.

3.4.8 *Ozone resistance.* Natural or synthetic rubber used externally on tubeless-tire valves only shall exhibit resistance to ozone cracking when tested as specified in 4.5.7.

3.4.9 *Crack resistance.* No cracks or cuts shall be visible in the rim-hole sealing area of type VII valves when tested as specified in 4.5.8.

3.4.10 *Metal characteristics of valve stems and spuds.*

3.4.10.1 *Type I, classes 1 and 7.* The metal valve stems shall be of such hardness as to prevent distortion when subjected to normal mounting and servicing methods. Valve stems of valves TR–135, TR–150, and TR–150CW shall permit hand bending without cracking when bent as specified in 4.5.10.1.

3.4.10.2 *Type I, class 8; type II, classes 1 through 7, and type III.* The metal valve stems and spuds shall not be distorted when subjected to normal mounting and servicing methods.

3.4.10.3 *Type I, classes 2, 3, 4, 5, 6, 9, and 10.* The metal, convertible valve stems shall withstand bending, without cracking or breaking, when tested as specified in 4.5.10.2.

3.4.10.4 *Type I, classes 11 and 12.* The metal, convertible valve stems shall withstand bending, without cracking or breaking, when tested as specified in 4.5.10.3.

3.5 **Dimensions.** Dimensions of valve assemblies, caps, cores, and spuds shall be as specified in MS51357 through MS51378. Type I, classes 11 and 12 dimensions shall conform to those shown in the Year Book of The Tire and Rim Association, Inc.

3.6 **Applicability of fittings.** Applicability of valve cores and caps shall be as specified in MS51375 through MS51378. Applicability of valve spuds shall be as specified in MS51374.

3.7 **Marking.** Each valve and valve cap

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shall be stamped, molded or otherwise permanently and legibly marked in accordance with the applicable MS standard and Military Standard MIL–STD–130.

3.8 **Workmanship.** The workmanship shall be of a quality necessary to produce valves free from any defects which affect their appearance or impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.

4.1.1 *Contractor inspection.* The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other inspection facilities and services acceptable to the Governments. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 **Preproduction inspection.** A preproduction sample of eight of each type specified, shall be furnished for preproduction inspection. The samples shall be representative of the units proposed to be furnished under the contract (see 3.1). Preproduction inspection shall be conducted by the Government at a place designated by the Government, and shall consist of examination for the defects listed in 4.3.2.2, and testing as specified in 4.5 (table II).

4.2.1 *Type I, classes 1 through 8, and type II, class 1 valves.* All type I, classes 1 through 8, and type II, class 1 valves supplied for preproduction testing shall be accompanied by samples of the natural or synthetic rubber used on their bases and stems. Samples shall be approximately 1 inch square by a minimum of ¼ inch thick and shall be certified by the supplier as being exactly the same as, or as having come from the same batch that was used on the valves submitted for preproduction testing.

4.2.2 *Failure.* Failure of a preproduction sample to pass any of the inspection specified herein may be cause for the Government to refuse to conduct additional inspection until it has been proved to the satisfaction of the Government that the faults revealed by the inspection have been corrected.

4.3 Acceptance inspection.

4.3.1 *Sampling for valves, valve spuds, valve caps, and valve cores.*

4.3.1.1 *Lot formation.* Unless otherwise specified, a lot shall consist of all valves, valve spuds, valve caps, and valve cores, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.3.1.2 *Sampling for acceptance examination.* Samples for acceptance examination shall be selected in accordance with Standard MIL–STD–105 (2.1).

4.3.1.3 *Sampling for acceptance testing.* Samples for acceptance testing shall be selected in accordance with inspection level L6 of the appendix to Standard MIL–STD–105 (see 2.1).

4.3.2 *Examination inspection for valves.*

4.3.2.1 *Acceptable quality level.* Each valve, and each valve cap and valve core (as applicable) selected in accordance with 4.3.1.2, shall be examined for conformance to the following acceptable quality levels (AQL) on the basis of percent defective:

Classification	*AQL*
Major	1.0
Minor	2.5

4.3.2.2 *Classification of defects.* For examination purposes, defects shall be classified as follows:

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Categories	Defects	Method of inspection
Critical	None defined	
Major	AQL 1.0 percent	
101	Incorrect base preparation (see 3.2.2.1)	Visual.
102	Incomplete assembly (see 3.3.1)	Visual.
103	Incorrect valve-cap construction (see 3.3.3).	Visual.
104	Incorrect valve-core constriction (see 3.3.4).	Visual.
105	Metal characteristics nonconformance (see 3.4.10).	Visual.
106	Incorrect dimensions (see 3.5)	Gage.
Minor	AQL 2.5 percent	
201	Marking nonconformance (see 3.7)	Visual.

4.3.3 *Testing inspection for valves.* Samples selected in accordance with 4.3.1.3 shall be subjected to the applicable tests specified in table III, using an AQL of 6.5 on the basis of percent defective. All type I, classes 1 through 8, and type II, class 1 valves supplied for acceptance testing shall be accompanied by samples of the natural or synthetic rubber used on their bases or stems. Samples shall be approximately 1 inch square by a minimum of ¼ inch thick and shall be certified by the supplier as being exactly the same as, or having come from the same batch that was used on the valves being acceptance tested.

TABLE II. *Classification of tests.*

Paragraph No.	Test	Preproduction	Acceptance	Control
4.5.1.2.1	Higher-temp. leakage (other than type II, class 1)	X	.....	X
4.5.1.2.2	Low-temp. leakage (other than type II, class 1)	X	.....	X
4.5.2.1	High-temp. leakage (type II, class 1 only)	X	.....	X
4.5.2.2	Low-temp. leakage (type II, class 1 only)	X	.....	X
4.5.3	Pull	X	X	
4.5.4	Durometer hardness	X	X	
4.5.5	Closure	X	X	
4.5.6	Valve-core leakage*	X	X	
4.5.7	Ozone resistance	X		
4.5.8	Crack resistance	X	X	
4.5.9	Valve-cap leakage**	X	X	
4.5.10	Bendability	X	X	

* Valve cores, when purchased separately, shall be acceptance tested as specified in 4.5.6.
** Valve caps, when purchased separately, shall be acceptance tested as specified in 4.5.9.

4.4 Control tests.

4.4.1 *Sampling for control tests.* Samples shall be selected at the rate of 10 of each 1000 units produced on each contract except that not more than 20 nor less than 10 units shall be selected in any 30 day period. The first sample shall be taken at or near the start of production. Samples shall be subjected to the applicable tests specified in tables II and III.

4.4.2 *Failure.* Failure of a control-test sample to pass any specified examination or test may be cause for the Government to refuse to accept subsequent lots until it has

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TABLE III. *Applicability of tests.*

	Type																		
	Class																		
	I												II						
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3	4	5	6	7
High-temperature leakage (other than type II, class 1) (par. 4.5.1.2.1)	X	X	X	X	X	X	X	X	X	X	X	X	..	X	X	X	X	X	X
Low-temperature leakage (other than type II, class 1) (par. 4.5.1.2.2)	X	X	X	X	X	X	X	X	X	X	X	X	..	X	X	X	X	X	X
High-temperature leakage (type II, class 1) (par. 4.5.2.1)	..	..	..	..	..	..	..	..	..	..	..	..	X						
Low-temperature leakage (type II, class 1) (par. 4.5.2.2)	..	..	..	..	..	..	..	..	..	..	..	..	X						
Pull (par. 4.5.3)	X	X	X	X	X	X	X	X	..	..	..	..							
Durometer (par. 4.5.4)	X	X	X	X	X	X	X	X	..	..	..	..	X						
Closure (par. 4.5.5)	..	..	..	..	..	..	X	X	..	..									
Valve-core leakage (par. 4.5.6)	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Ozone resistance (par. 4.5.7)	..	..	..	..	..	..	..	..	..	..	..	..	X						
Crack resistance (par. 4.5.8)	..	..	..	..	..	..	..	..	..	..	..	..	X						
Valve-cap leakage (par. 4.5.9)	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Bendability (par. 4.5.10)	X	X	X	X	X	X	X	..	X	X	X	X							

been proved to the satisfaction of the Government that the faults revealed by the tests have been corrected.

4.5 Tests.

4.5.1 *Temperature-range performance (all except type II, class 1).*

4.5.1.1 *Apparatus.* Test apparatus shall include a test fixture into which valves to be tested can be installed. Suitable equipment shall be available to determine the air pressure and air flow through the valve assembly.

4.5.1.2 *Procedure.* The valve assemblies shall be installed in the test fixture and the valve cores hand-tightened (see 3.4.4) in the valve stems.

4.5.1.2.1 *High-temperature leakage.* The test fixture, with valve assemblies installed and with a pressure of 100 psi applied, shall be heat aged in a circulating air oven at a temperature of 212° F. After 24 hours of aging, the assembly shall be placed in a water bath of 200° F. and soaked for 10 to 15 minutes. The valves, while still immersed, and with pressure applied, shall be checked for conformance to the leakage requirements of 3.4.5.1.

4.5.1.2.2 *Low-temperature leakage.* The test fixture, with previously untested valve assemblies installed and with a pressure of 100 psi applied, shall be submerged in water, the temperature of which is 75° ± 5° F., and the assembly checked for air leakage for conformance to 3.4.4. The test fixture with valve assemblies shall then be placed in a Dewar flask (or equivalent) with an alcohol bath at minus 65° F. and conditioned for 30 minutes with a contained air pressure of 100 psi. After checking the valve assemblies for leaks, the test fixture and valves shall be removed from the bath, the plunger pins depressed, and the fixture returned to the alcohol bath for a five minute conditioning period. Observations shall then be made to determine conformance to the leakage requirements of 3.4.5.2.

4.5.2 *Temperature range performance (type II, class 1 only).*

4.5.2.1 *High-temperature leakage.*

4.5.2.1.1 *Apparatus.* A size 5–K rim of nominal thickness shall be used for test purposes. It shall have bur-free valve holes of 0.625 ± .007 inch in diameter for the testing of large size and of 0.453 ± .005 inch in

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diameter for the testing of the smaller size, type II, class 5 (snap-in) replacement valves. Standard, metal valve caps with swivel rubber seals and metal guide-ring jigs shall be used.

4.5.2.1.2 *Procedure.* Valves to be tested shall be inserted into the holes in the rim, using a soap solution to lubricate the mounting. The valves shall then be subjected to 24 pounds per square inch inflation pressure at room temperature. The assembly shall be placed in an air oven, having a temperature of 212° F., for 72 hours. At the end of every 8 hours (of the 72), the valves (under pressure) shall be immersed in water and flexed. This check for leakage shall be made by bending the valves 25° by hand and rotating them slowly one complete revolution. This operation shall be repeated at ½-hour intervals for a total of five times. Inflation pressure shall be checked before each 8-hour flexing operation. Valves shall be returned to oven without adjusting air pressure. A check for leakage of air shall be made before, during, and after flexing to determine conformance to 3.4.6.1.

4.5.2.2 *Low-temperature leakage.*

4.5.2.2.1 *Apparatus.* The apparatus described in 4.5.2.1.1 shall be used.

4.5.2.2.2 *Procedure.* Valves to be tested shall be inserted into the holes in the rim using a soap solution to lubricate the mounting. Mounted valves shall stand for 24 hours at room temperature to dry prior to test. Assembly shall then be placed in a cold chamber, the temperature of which shall be maintained at minus 65° ± 2° F. While in the cold chamber, and with inflation pressure maintained on them at 24 psi, valves shall be soaked in an alcohol bath for 24 hours. Valves shall then be checked for leakage by bending them by hand 25° from the vertical and slowly rotating them one complete revolution. While still being soaked, the rotating shall be repeated at half-hour intervals for a total of five times, beginning 30 minutes after completion of the first test, during and after which times the valves shall be examined for conformance to 3.4.6.2.

4.5.3 *Pull test for separation of rubber ferrule from metal valve stem (type I, classes 1 through 8 valves).* The valve base shall be clamped in a fixture with an adapter ring diameter as specified in table I. Cap thread or body thread shall be attached to threaded adapter in the pull mechanism. The assembly shall be placed in a tensile machine and a direct pull shall be made at the rate of 5 inches per minute until destruction. Pull values shall conform to those specified in table I (see 3.4.1).

4.5.4 *Durometer hardness.* To determine conformance to 3.4.2, the valve-rubber compound (see 4.2.1) shall be subjected to the test for durometer hardness specified in method 3021 of Standard FEDERAL TEST METHOD STD. NO. 601, using a type A Shore durometer.

4.5.5 *Closure.* To determine conformance to 3.4.3, the rubber-base stem of the valve being tested shall be cut to expose the degree of closure in the valve-stem base.

4.5.6 *Valve-core leakage.*

4.5.6.1 *Apparatus.* The test apparatus shall include a manifold having tank valve stems into which valve cores can be fitted. Equipment shall be available to determine air pressure.

4.5.6.2 *Procedure.* Valve cores shall be inserted into the tank valve stems of the test manifold. The valve cores shall then be screwed into the valve stems and tightened to 2 pound-inches of torque. The test manifold, with valve cores installed, and with a pressure of 75 psi applied, shall be immersed in water for 1 minute. The test manifold shall then be removed from the water and the valves visually examined to determine their conformance to 3.4.4.

4.5.7 *Ozone resistance.*

4.5.7.1 *Apparatus.* A flat base plate shall be used drilled with holes .618 inch to .633 inch or .448 inch to .458 inch as required in diameter and spaced 1.50 inch center to center. Edges of holes in plate shall be chamfered per Tire and Rim Association,

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Inc. specifications. The base plate shall be ⅛-inch thick and approximately 4 inches wide. Base-plate length shall depend upon the number of valves to be tested. Another flat plate, ⅛-inch thick and 1½ inches wide and the same length as the base flat plate, shall be used to bend the valves to a specified angle by positioning the plate between the valves at the threaded portions. Small holes shall be drilled along the base and a wire shall be used to hold the valves to the specified angle. Valves shall be at least 72 hours old before testing.

4.5.7.2 *Procedure.* Valves shall be lubricated with soap and mounted in the plate. Valves shall be washed with cold tap water for a minimum of 15 minutes to remove the lubricant. Valves shall then be bent to a 10° angle and aged for 24 hours in a dark cabinet at room temperature. Valves, while still bent at the 10° angle, shall be placed in an ozone chamber in which the ozone concentration is maintained at 50 ± 5 parts of ozone to 100,000,000 parts of air by volume. The air, with ozone concentration, shall be circulated throughout the test at a temperature of 100° ± 5° F. Valves shall remain in this ozone chamber for 168 hours before removal for inspection under 9-power magnification to determine conformance to 3.4.8.

4.5.8 *Crack resistance.* The straight length of the rim hole sealing area shall be stretched to a ratio of 2 to 1 to elongate the section sufficiently to permit visual examination to determine conformance to 3.4.9.

4.5.9 *Valve-cap leakage.* Valve caps to be tested shall be finger-tightened on valve stems having no valve cores. A pressure of 100 psi shall then be supplied to the valve interior. While still under pressure the valve shall be immersed in water for thirty seconds to determine conformance to 3.4.7.

4.5.10 *Bendability of valve stems.*

4.5.10.1 *Type I, classes 1 and 7.* Hand-bendable valve stems shall be held solid in a fixture and each stem then hand bent to an angle not to exceed 70°. The valves shall then be examined to determine conformance to 3.4.10.1.

4.5.10.2 *Type I, classes 2 through 6, 9, and 10.* The metal valve stems shall be placed in standard valve-bending tools and bent a full 86° angle at the valve-base end and bent a full 90° angle at not less than 1½ inches from the valve-cap end of each valve stem. The valve stems shall then be examined to determine conformance to 3.4.10.3.

4.5.10.3 *Type I, classes 11 and 12.* The metal convertible valve stems shall be placed in standard valve-bending tools and bent a full 86° angle at the valve-base end, and bent a full 90° angle at not less than 2 inches from the valve-cap end of each valve stem. The valve stems shall then be examined to determine conformance to 3.4.10.4.

4.5.11 *Sampling for packaging and packing.*

4.5.11.1 *Lot formation.* A lot shall consist of all packs prepared for shipment in accordance with one level (see 5.2), from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.5.11.1.1 *Sampling for acceptance examination.* Samples for acceptance examination shall be selected as specified in Standard MIL–STD–105.

4.5.11.1.2 *Sampling for acceptance testing.* One package and one complete pack shall be selected for acceptance testing.

4.5.11.2 *Examination inspection for packaging and packing.*

4.5.11.2.1 *Acceptable quality level.* Samples selected in accordance with 4.5.11.1.1 shall be examined for conformance to the following acceptable quality levels (AQL), on the basis of percent defective:

Classification	*AQL*
Major	1.0
Minor	2.5

4.5.11.2.2 *Classification of defects.* For examination purposes, defects shall be classified as follows:

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Categories	Defects	Method of inspection
Critical	None defined	
Major	AQL 1.0 percent	
101	Parts improperly cleaned (see 5.1.1.1)	Visual.
102	Improper carton (see 5.1.1.2)	Visual.
103	Valve assemblies with rubber bases, packaged with rubber base distorted (see 5.1.1.2).	Visual.
104	Parts not properly cushioned (see 5.1.1.2).	Visual.
105	Improper cushioning material used (see 5.1.1.2).	Visual.
106	Improper quantity of pieces in unit carton (see 5.1.1.2).	Count.
107	Weight of unit carton exceeds 10 pounds (see 5.1.1.2).	Weighing scale.
108	Illegible marking (see 5.3)	Visual.
109	Incorrect marking (see 5.3)	Visual.
Minor	AQL 2.5 percent.	
201	Improper closure of strapping of shipping container (see 5.2.1–5.2.2).	Visual.

4.5.11.3 *Testing inspection for packaging and packing.* When level A or B is specified (see 6.2), the samples selected in accordance with 4.5.11.1.2 shall be subjected to the test specified in 4.6.1 and 4.6.2.

4.5.11.3.1 *Failure.* Failure of the package or pack to pass any specified test, may be cause for the Government to refuse to accept the lot until it has been proved to the satisfaction of the Government that the faults revealed by the tests have been corrected.

4.6 Packaging and packing test procedures.

4.6.1 *Packaging test.* To determine conformance to 5.1.1.2, the unit package shall be subjected to the tests specified for method III of Specification MIL–P–116.

4.6.2 *Packing test.* To determine conformance to 5.2.1, the shipping container shall be subjected to the interior packing test of Specification JAN–P–100. To determine conformance to 5.2.2, the same test shall be performed, except the height of the drop shall be 75 percent of the height of the drop specified therein.

5. PREPARATION FOR DELIVERY

5.1 Cleaning and packaging. Cleaning and packaging shall be level A or C as specified (see 6.2).

5.1.1 *Level A.*

5.1.1.1 *Cleaning.* Exposed, unpainted surfaces shall be cleaned by method C-1 of Specification MIL–P–116 and dried by any applicable procedure specified therein.

5.1.1.2 *Packaging.* Immediately after cleaning and drying, valve assemblies shall be packaged in a close-fitting box, conforming to type I or II, class 2, of Specification PPP–B–636. Care shall be exercised to assure that valve assemblies with rubber bases are packaged in such a manner that the rubber bases are not distorted within the package. Cushioning material shall be used to prevent free movement of the valve assemblies within the box. Cushioning material shall consist of paperboard cushioning material conforming to the most applicable type and class of Specification PPP–P–291, or cellulosic cushioning material conforming to the most applicable type and class of Specification PPP–C–843. Unit package shall contain 20 to 50 pieces, provided gross weight does not exceed 10 pounds. When tested as specified in 4.6.1 the package shall meet the requirements of method III of Specification MIL–P–116.

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5.1.2 *Level C.* Cleaning and packaging shall be in accordance with the manufacturer's commercial practice.

5.2 **Packing.** Packing shall be level A, B or C as specified (see 6.2).

5.2.1 *Level A.* Unit packages as described in 5.1 shall be packed in a wood-cleated plywood box, nailed wood box, wood fiberboard-lined box, wire-bound wood box, or fiberboard wood-cleated box, conforming to the requirements of Specification PPP–B–601 overseas type, PPP–B–621 class 2, MIL–B–138, PPP–B–585 class 3, or PPP–B–591 overseas type, respectively. Gross weight shall not exceed 200 pounds. When tested as specified in 4.6.2 the pack shall meet the requirements of the interior packing test of Specification JAN–P–100.

5.2.2 *Level B.* Unit packages as described in 5.1 shall be packed in a wood-cleated plywood box, nailed wood box, wirebound wood box, or fiberboard wood-cleated box conforming to the requirements of Specification PPP–B–601, domestic type, PPP–B–621, class 1, PPP–B–585, class 1 or PPP–B–591, domestic type, respectively. Gross weight shall not exceed 200 pounds. When tested as specified in 4.6.2, the pack shall meet the requirements of the interior packing test of Specification JAN–P–100, except the height of the drop shall be 75 percent of the height of the drop specified therein.

5.2.3 *Level C.* Unit packages, as described in 5.1, shall be packed in accordance with Consolidated Freight Classification, National Motor Freight Classification, or other common carrier regulations applicable to the mode of shipment.

5.3 **Marking.** In addition to any special marking required by the contract or order, the unit pack and shipping containers shall be marked in accordance with the requirements of Standard MIL–STD–129.

6. NOTES

6.1 **Intended use.** The pneumatic-tire valve and valve spuds covered by this specification are intended primarily for the following uses:

(a) Type I, classes 1 through 8 valves are used for the replacement of faulty or damaged, rubber-covered, air-water, or rubber-base, metal-stem, tube valves.

(b) Type I, classes 9 through 12 valves are of the screw-on design and are used for the repair of standard and large-bore tube valves.

(c) Type II valves are of either snap-in or clamp-in design and are used for the replacement of defective or damaged tubeless valves of corresponding design.

(d) Type III valve spuds are used for the replacement of defective spuds of corresponding design or as an assembly, with applicable stems, to form repair or replacement assemblies as specified in MS51374.

6.2 **Ordering data.** Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Type and class of valve, valve spud, valve cap, or valve core required (see 1.2).

(c) Tire and Rim Association, Inc. valve number and description of valve-stem bend, if any (see 6.3).

(d) Type of rubber cover for snap-in valve if different (see 3.2.2).

(e) Where preproduction samples are to be submitted for inspection (see 3.1 and 4.2).

(f) Selection of level of packaging (see 5.1).

6.3 **Identification of valves by types.** Identification of valve assemblies by Tire and Rim Association, Inc. valve number, as applied to types and classes, shall be as indicated in Military Standards MS51357 through MS51373.

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